

Natural Language Processing and Large Language Models

Corso di Laurea Magistrale in Ingegneria Informatica



Project Assignment

Nicola Capuano and Antonio Greco

DIEM – University of Salerno





Problem Statement

DIEM website contains a large amount of information distributed across **multiple webpages and documents**...

- Core website sections: Home, Department, Teaching, Research, Third Mission, International
- Faculty members information
- Information on degree and PhD programs
- Course and exam timetables
- Regulations

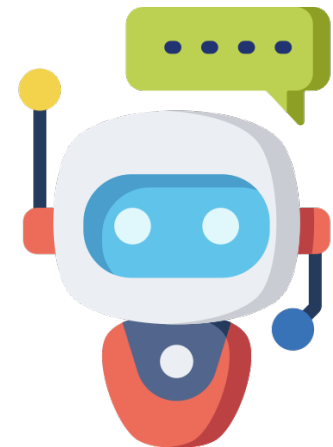
As a result, users need to manually navigate **multiple pages** to find specific information...

Project Goal

Build an **LLM-powered chatbot** capable of providing **accurate, relevant, and conversational** answers to user questions about DIEM services, academic activities, facilities, and official information.

The chatbot should:

- Answer questions related to DIEM website content
- Retrieve updated information from department pages
- Handle multilingual queries (**optional**)
- Detect out-of-domain questions
- Avoid hallucinations





Sample Questions

Prospective Students

- What degree programs are offered by DIEM?
- I got a score of 18 on my TOLC. Can I enroll?
- What are the admission requirements for the Master's Degree in Information Engineering for Digital Medicine?

Current Students

- What is the syllabus of the Software Engineering course?
- What are Professor Capuano's office hours?
- Where is Room 126 located?
- My average grade is 28.8. What is the highest final grade I can obtain?



Sample Questions

International Students:

- Who is responsible for internationalization at DIEM?
- What international mobility opportunities are available?

Others (external users, companies, researchers, prospective collaborators, and visitors seeking general information)

- Where is DIEM located?
- What research areas are active at DIEM?
- Which laboratories are available at DIEM?
- Which equipment is available in the Robotics Laboratory?
- Who are the members of the Joint Students-Teachers Commission?



Knowledge Scope

The chatbot is expected to answer questions based only on information available from the following **sources**:

- Pages under the <https://www.diem.unisa.it/> domain
- Pages under <https://docenti.unisa.it/> related to DIEM faculty members
- Pages under <https://corsi.unisa.it/> related to courses offered by DIEM
- PDF documents referenced by these pages, such as regulations, guidelines, and official notices
- Course and exam timetables of DIEM courses available from <https://easycourse.unisa.it/> (optional)

External links should not be considered



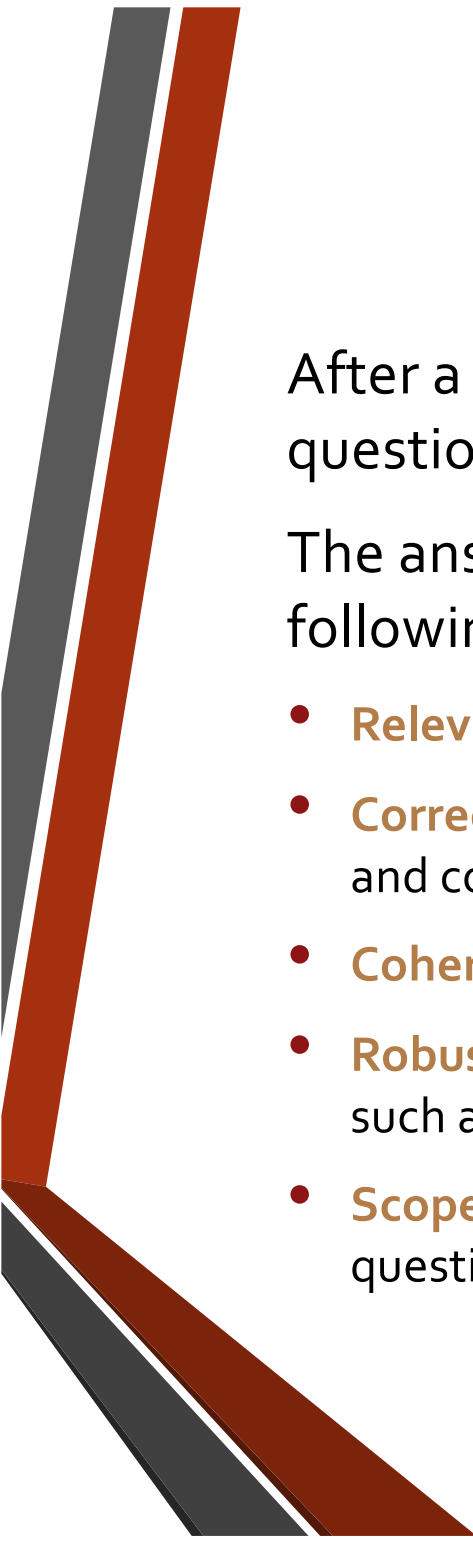
Procedure

- Each project must be developed by a **group of three or four students**
- The **group composition** must be communicated in the following **online sheet** (within one week)
- Each group must submit:
 - The **chatbot source**
 - A **final report** describing the proposed solution in detail
- The submission must be sent to the teachers (by e-mail) at least **three working days before the exam date**



Procedure

- During the **first part of the exam**, each submitted project will be **evaluated**
- In case of **positive evaluation**, group members are enabled to take the second part of the exam, focused on theory
 - This may happen either in the **same exam session** or in a **subsequent one (within one year)**
- The final grade will be computed as the **average** between the **project score** and the **theory score**



Evaluation Criteria

After a brief **demo** (up to 5 minutes), the instructors will question the chatbot using a predefined set of questions

The answers will be **qualitatively evaluated** according to the following criteria:

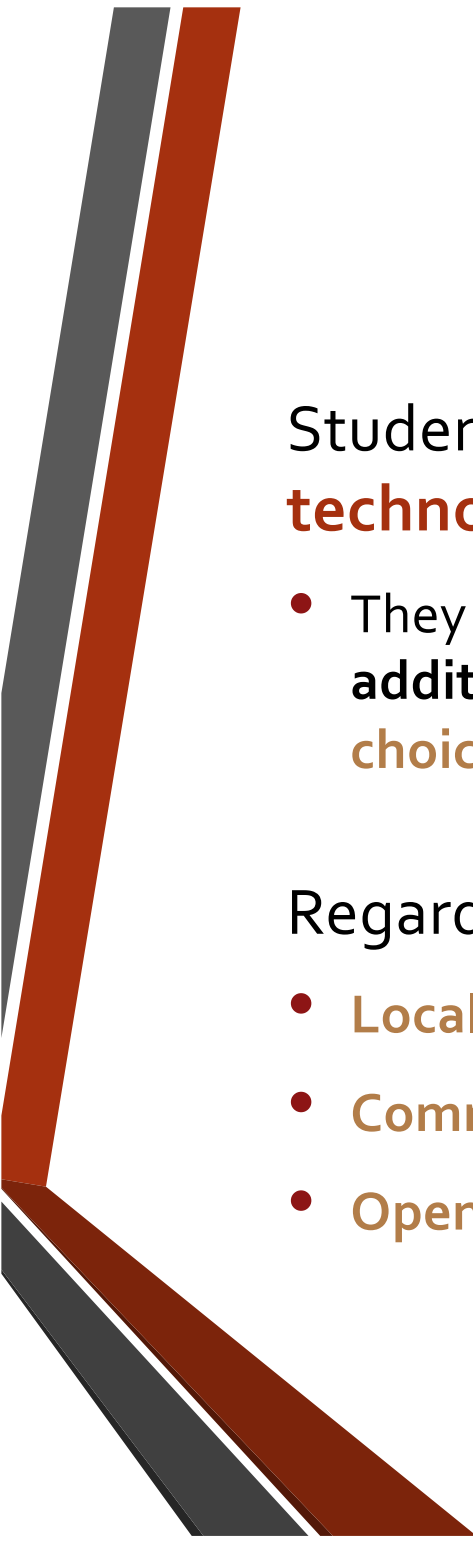
- **Relevance**: evaluates whether the generated answer addresses the query
- **Correctness**: evaluates whether the generated answer is factually accurate and consistent with the source information
- **Coherence**: checks the logical flow and internal consistency of the answer
- **Robustness**: evaluates resistance to adversarial or misleading prompts, such as “Are you sure?”
- **Scope Awareness**: evaluates the ability to recognize out-of-context questions, such as “Who is the king of Spain?”



Evaluation Criteria

The final **project score** will be based on:

- **Chatbot Performance – 70%**
 - Qualitative indicators evaluated on the chatbot's answers
- **System Design and Documentation – 30%**
 - System architecture, as described in the final report
 - Overall quality of the final report
 - User experience of the developed chatbot
 - Clarity and effectiveness of the live demo



Tools and Technologies

Students are free to choose **any tools, frameworks, and technologies** for developing the chatbot

- They may use solutions **covered during the course**, as well as **additional tools** not explicitly discussed, provided that **their choices are properly justified** in the final report

Regarding **LLMs**, students are also free to choose among:

- **Locally hosted models**
- **Commercial hosted APIs**
- **Open-source models**

Natural Language Processing and Large Language Models

Corso di Laurea Magistrale in Ingegneria Informatica



Project Assignment

Nicola Capuano and Antonio Greco

DIEM – University of Salerno

